

Chapter 125 — Genital and Lower Urinary Tract Trauma

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- Lower urinary tract injuries are **rarely life-threatening but highly morbid if mismanaged** — long-term disability follows error, not the injury itself.
- Outcomes rest on **appropriate imaging plus correct timing and approach of intervention**.
- The central acute decision is **immediate reconstruction versus deliberate delay: testicular rupture → immediate reconstruction; pelvic fracture urethral injury → delayed reconstruction**.
- Correct acute management is what **makes delayed reconstruction feasible** — large-bore suprapubic tube at the time of severe urethral injury, urethroplasty later.
- Imaging has advanced such that **site localisation now dictates a more direct or minimally invasive approach: ultrasonography** routinely for scrotal trauma and increasingly to confirm penile fracture; **high-resolution computed tomography (CT) cystography** to pinpoint bladder rupture, enabling laparoscopic or robotic repair; **retrograde urethrography** to diagnose pelvic fracture urethral injury and separate those suitable for primary realignment from those better managed by maintaining a suprapubic tube.

INJURIES OF THE EXTERNAL GENITALIA

Penis

- Genital trauma is uncommon largely because of the **mobility of the penis and scrotum**, which allows displacement rather than absorption of force.
- Blunt phallic trauma is **of concern chiefly in the erect state**, where fracture of the tunica albuginea may occur.
- **Prompt surgical reconstruction of most penile injuries yields acceptable cosmetic and functional results.**

Penile Fracture

Aetiology and anatomical basis

- Definition: **disruption of the tunica albuginea with rupture of the corpus cavernosum**.
- The tunica albuginea is **bilaminar — inner circular layer, outer longitudinal layer** — composed of collagen and elastin.
- The **outer longitudinal layer determines tunical strength and thickness**, which varies along the shaft and is **thinnest ventrolaterally**.
- Mechanism: abnormal bending of the erect penis causes an **abrupt rise in intracavernosal pressure exceeding tunical tensile strength**, producing a **transverse laceration**.
- Laterality: **usually unilateral; bilateral corporal tears in approximately 10%. Bilateral corporal injury carries a higher association with urethral injury** — treat bilaterality as a prompt to interrogate the urethra.
- Site: the typical tear is a **transverse laceration of the proximal to mid shaft, distal to the suspensory ligament**; coital injuries are typically **ventral or lateral**, corresponding to the thinnest region of tunica.
- **Sexual intercourse is the commonest cause**, with the **rear-entry ("doggy style") position most frequently implicated**. Also described with **masturbation, rolling onto or falling onto the erect penis**.
- Reported to occur more often in **"stressful situations" such as extramarital intercourse** — relevant because it predicts **delayed presentation**.

- In the Middle East, **self-inflicted fracture predominates via "taqaandan"** — forcible bending of the erect penis during masturbation or to achieve rapid detumescence after coitus. **Cultural factors account for the geographical variation in aetiology.**
- **Collagenase clostridium histolyticum (CCH) injection for Peyronie disease is an increasingly recognised cause.** Pooled trial safety data show **corporal rupture to be an uncommon but real complication**, with a further group of unconfirmed suspected ruptures.
- Single-centre experience: **corporal rupture occurred at a higher rate than trial data suggested, and the majority arose after the 14-day observation window had closed — the standard observation period does not capture most events.**
- Most CCH-related ruptures still occur **during sexual activity**, though cases attributed to **nocturnal tumescence** are reported.

Diagnosis and imaging

- **Diagnosis is clinical:** history and physical examination are reliable in most cases (**American Urological Association [AUA] Urotrauma Guidelines, Statement 26**).
- Classic history: **a cracking or popping sound as the tunica tears → pain → rapid detumescence → discoloration and swelling of the shaft.**
- The **commonest findings are haematoma (essentially universal) and detumescence.**
- **Buck's fascia determines the pattern of haematoma — the key anatomical discriminator: intact → haematoma confined between skin and tunica → "eggplant deformity"; disrupted → haematoma tracks to scrotum, perineum and suprapubic region.**
- The **swollen, ecchymotic phallus deviates away from the side of the tunical tear**, owing to haematoma mass effect. The **fracture line in the tunica may be palpable.**
- **Fear and embarrassment commonly delay presentation.**
- Concomitant urethral injury differs sharply by region: **approximately 20% in the United States and Europe versus approximately 3% in Asia, the Middle East and the Mediterranean.** The explanation is **aetiological — intercourse trauma versus self-inflicted injury.**
- Most urethral injuries present with **gross haematuria, blood at the meatus, or inability to void — but absence of these findings does not exclude urethral injury.**
- **Where urethral injury is suspected, urethral evaluation is compulsory.** Preoperative retrograde urethrography may be considered but is **time-consuming and can be inaccurate.**
- **Intraoperative flexible cystoscopy immediately before catheter placement at penile exploration is now the routine approach (AUA Urotrauma Guidelines, Statement 29).** The European Association of Urology (EAU) likewise recommends flexible cystourethroscopy in preference to retrograde urethrography, on the grounds of superior detection (**EAU Guidelines, 2020**).
- **Adjunctive imaging is unnecessary when the history and examination are typical — imaging is reserved for the equivocal case.**
- **Ultrasonography is the preferred imaging study for equivocal penile fracture (AUA Urotrauma Guidelines, Statement 28)** because it is **rapid, readily available, non-invasive, inexpensive and accurate.** Its two uses are **ruling out fracture where clinical suspicion is low and localising the tear, thereby guiding the choice of incision.**
- **Magnetic resonance imaging (MRI) is an accurate non-invasive alternative for demonstrating tunical disruption; AUA guidance recommends considering MRI when ultrasonography is equivocal.**

- **MRI's value lies in exclusion** — against intraoperative findings it showed **100% sensitivity and 100% negative predictive value**, with a **positive predictive value of 90.5%**: a negative MRI reliably excludes fracture; a positive one is less certain.
- **MRI is not used widely** because of **expense, limited availability and time requirements**.
Cavernosography is discouraged — time-consuming and unfamiliar to most urologists and radiologists.
- **"False fracture" presents with penile swelling and ecchymosis** and may reproduce the classic history, so the history alone does not confirm a corporal tear; the classic **"snap-pop" sound** is reported in only a minority of false fractures.
- **Physical examination may be inadequate to confirm or exclude a corporal tear** in these circumstances; surgical exploration or MRI should be considered.
- **Rupture of the dorsal penile artery or vein during intercourse is the other recognised mimic.**

Management

- **Suspected penile fracture warrants prompt exploration and surgical repair** (AUA Urotrauma Guidelines, Statement 27).
- The operation as a sequence: **exposure → evacuation of haematoma → identification of the tunical disruption → irrigation → closure of the tunical defect → repair of urethral injury where present.**
- **Ventral vertical penoscrotal incision is usually preferred**, because most fractures are ventral or lateral and this gives direct exposure. **Small lateral incisions** may suffice for a **localised haematoma** or a **palpable tunical defect**.
- **A distal circumcising incision risks ischaemia of the distal prepuce** in the uncircumcised patient — if required, **limited circumcision at the conclusion of the repair** should be strongly considered.
- **Close the tunical defect with interrupted 2-0 or 3-0 absorbable sutures.**
- **Avoid deep corporeal vascular ligation and excessive debridement of the underlying erectile tissue** — the tissue is delicate and over-treatment costs erectile function.
- **Artificial erection induced with saline or coloured dye aids localisation of the corporal laceration.**
- Associated urethral injury: **partial → oversew with fine absorbable suture over a urethral catheter; complete → debride, mobilise and repair tension-free over a catheter.**
- After urethral repair: **catheter maintained 2 to 3 weeks with broad-spectrum antibiotics, and 1 month of sexual abstinence.**
- Fracture after CCH injection: **management is controversial and there is no validated algorithm.** Survey practice was predominantly surgical, with **poor tissue quality and greater technical difficulty at repair** frequently reported.
- **The trend is towards conservative management, provided there is no concomitant urethral injury** — the presence of urethral injury remains an indication to operate.

Outcome and complications

- **Surgical reconstruction outperforms conservative management: faster recovery, reduced morbidity, lower complication rates and less long-term penile curvature** (AUA Urotrauma Guidelines, Statement 27).
- **Overall postoperative complication rate after repair is approximately 20%** — **palpable penile nodules, penile curvature, erectile dysfunction (ED), infection and pain** — which remains lower than the complication rate following observation.
- **A large surgical series reported preserved erectile function in nearly all operated patients: repair reliably protects erectile function.**

- **Delay of up to 7 days from injury does not adversely affect the outcome of repair** — the late presenter should still be explored.
- After CCH injection, conservatively managed suspected fractures showed **no difference in ED compared with those without fracture, and greater improvement in curvature**; separately, **no difference in sexual function, curvature or satisfaction between surveillance and surgery**. Conservative management is at least equivalent, and arguably preferable, in this setting.

Gunshot and Penetrating Injuries

Gunshot wounds

- **Gunshot wounds account for most penetrating genital injuries, and most require surgical exploration.**
- Treatment principles as a sequence: **immediate exploration → copious irrigation → excision of foreign matter → antibiotic prophylaxis → surgical tunical closure.**
- **Saline corporal injection assesses for corporal injury intraoperatively. Exploration must not be delayed for imaging.**
- **Penile gunshot wounds are almost never isolated** — nearly all victims have **significant associated abdominal, pelvic, lower limb, vascular or additional genitourinary injuries.**
- **Immediate reconstruction yields excellent cosmetic and functional results. Artificial erection confirms penile straightness and absence of residual tunical violation; plication may correct curvature arising from closure of a large corporal defect.**
- **Urethral injury complicates 15–50% of penile gunshot wounds** — a far higher burden than in blunt fracture.
- **Retrograde urethrography should be strongly considered in any penetrating penile injury**, particularly with **high-velocity missile injury, blood at the meatus, difficulty voiding, or a bullet trajectory passing near the urethra.**
- Intraoperative alternatives: **retrograde urethral injection of methylene blue or indigo carmine** identifies the injury site and the adequacy of closure; **pericatheter injection** serves the same purpose if a catheter is already in situ.
- Repair: **viable tissue → primary closure using standard urethroplasty principles**, with excellent reported results; **high-velocity or close-range shotgun injury with extensive tissue loss → staged repair with suprapubic urinary diversion**, particularly for **penile urethral injuries.**

Animal and human bites

- **Morbidity is determined by the severity of the initial wound**, not by the species alone. **Most victims are young males; dog bites are the commonest.**
- **Infective complications are unusual in animal bites because treatment is sought early** — the contrast with human bites turns entirely on presentation delay.
- Dog bite management as a sequence: **copious irrigation → debridement → immediate primary closure → prophylactic broad-spectrum antibiotic. Tetanus and rabies immunisation as appropriate.**
- **Prophylaxis must cover polymicrobial flora: amoxicillin–clavulanic acid, ceftriaxone, ampicillin–sulbactam, or clindamycin plus trimethoprim–sulfamethoxazole.**
- **Human bites produce contaminated wounds that often should NOT be closed primarily.** Victims typically present after substantial delay and more often with established gross infection.
- **Empirical antibiotic therapy for human bites: amoxicillin–clavulanic acid or moxifloxacin.**

Penile Amputation

- Rare, and usually the result of genital self-mutilation. Underlying psychopathology raises the probability of genital self-mutilation, so psychiatric consultation should be sought in every case.
- Three management options: primary reanastomosis, maturation of the distal stump, or delayed phalloplasty — primary reanastomosis is preferred.
- Microsurgical repair is the preferred method — reconstruction of the urethra and corporal bodies with microsurgical repair of the dorsal penile vessels and nerves gives markedly superior results.
- Transfer to a centre with microsurgical capability should be attempted; where unavailable, macroscopic anastomosis of urethra and corpora still achieves good erectile results, at the cost of compromised sensation and skin loss.
- Preservation of the severed part uses the "double bag" technique: rinse the distal penis in saline → wrap in saline-soaked gauze → seal in a sterile plastic bag → place that bag inside an outer bag containing ice or slush (AUA Urotrauma Guidelines, Statement 32).
- The inner bag exists to prevent direct ice contact — prolonged direct contact causes hypothermic injury to the amputated segment.
- No significant association between ischaemia time and outcome has been found: a long ischaemia time is not by itself a reason to abandon replantation.
- If the severed part is unavailable, formalise the stump: close the corpora and spatulate the urethral neomeatus, as in partial penectomy for malignant disease.
- Outcome data for both macrovascular and microvascular reconstruction are limited and of poor quality, but microvascular reconstruction of the dorsal arteries, vein and nerves remains the preferred repair.
- More than 50% of men achieve erection after either microvascular reanastomosis or macroscopic replantation — the difference between techniques lies in the secondary outcomes.
- Urethral stricture, skin loss and sensory abnormality are all less common after microvascular repair. Return of skin sensation: 52% after microvascular reconstruction versus 14% after simple replantation, although only 42% of the microvascular group actually had a nerve repair performed, so the advantage of formal nerve repair is likely understated.
- Penile skin necrosis, sometimes complete, is a troublesome and frequent problem, less common with microsurgical repair. Penile skin has a blood supply independent of the corporal bodies, so without repair of the superficial vascular structures the skin behaves essentially as a free graft.
- Split-thickness skin grafts are applied once native skin becomes necrotic, typically after several weeks of observation, resuscitation and debridement.
- Reported adjuncts: hyperbaric oxygen to promote healing, and medicinal leeches after macroreplantation to augment venous outflow and reduce oedema.
- Total penile loss is addressed by total phallic reconstruction, classically the radial artery-based forearm free flap.

KEY POINTS: STEP-BY-STEP APPROACH TO PENILE REATTACHMENT

- Suprapubic cystostomy
- Debridement of non-viable tissue
- Closure of the tunica albuginea with 2-0 absorbable suture
- Urethral closure over a catheter with 4-0 absorbable suture
- Minimal dissection along the neurovascular bundle to identify severed vessels and nerves

- **Microscopic anastomosis of the dorsal artery with 11-0 nylon; dorsal vein repair with 9-0 nylon; epineural repair of the dorsal nerve with 10-0 nylon**
- **Skin coverage**

Zipper Injuries

- **Predominantly paediatric and predominantly penile.**
- First-line manoeuvre as a sequence: **penile block → lubricate the zipper slider and adjacent skin with mineral oil → a single attempt to unzip and disentangle the skin.**
- Mechanical release where that fails: **perpendicular cuts in the cloth between each zipper tooth**, removing the zipper's lateral support so the device falls apart; **cut the median bar (the diamond-shaped connection) of the slider with a bone cutter**, separating the upper and lower shields so the whole zipper disassembles; **insert a screwdriver between the shields and twist**; or **cut the anterior shield with a wire cutter.**
- **Some children require more than local anaesthesia or sedation — circumcision or elliptical skin excision under general anaesthesia is the fallback.**

Strangulation Injuries

- **Accidental injury from thread, hair or rubber bands occurs in children — but child abuse must be actively considered.**
- **Any child with unexplained penile swelling, erythema or difficulty voiding must be examined closely for a concealed strangulating hair or string.**
- **In adults the constricting object is usually applied for sexual pleasure or to prolong erection.**
- Pathophysiology in sequence: **reduced blood flow → oedema → ischaemia; gangrene and urethral injury develop in delayed presentations.**
- **Emergency treatment is decompression, to restore blood flow and permit micturition. Thread, hair and rubber bands are incised promptly.**
- Solid constricting device, as an escalating sequence: **lubricate shaft and object and attempt direct removal — distal oedema is what defeats this; wrap a string or latex tourniquet around the distal shaft to reduce swelling; then the string technique — pass thick silk suture or umbilical tape proximally beneath the object → wind it tightly around the penis distally towards the glans → grasp the proximal tag → unwind from the proximal end, which advances the object distally.**
- **Glanular puncture with needle or blade releases dark trapped blood and improves the success of the string method.**
- Material determines the tool: **plastic devices → scalpel or oscillating cast saw; heavy iron or steel → ordinary hospital equipment (ring cutters, bolt cutters, dental and orthopaedic drills, rotary tools) is often inadequate, and industrial drills, steel saws, hacksaws, sabre saws and high-speed electric drills, or occasionally fire service and emergency medical services equipment, are required.**
- **Protect the phallus from thermal injury, sparks and the blade with tongue depressors, sponges or malleable retractors, using continuous saline irrigation for cooling. These procedures are best performed in theatre under anaesthesia.**
- **If decompression is delayed and the patient is distended and unable to void, place a suprapubic catheter.**
- **Outcomes are generally favourable with device removal alone, but be prepared for reconstruction — skin grafting — if strangulation has caused skin necrosis.**

Testis

Aetiology

- The testis is relatively protected by three mechanisms: **mobility of the scrotum, reflexive cremasteric contraction, and the tough fibrous tunica albuginea.**
- Blunt force may nonetheless produce **rupture of the tunica albuginea, contusion, haematoma, dislocation, or torsion.** Blunt mechanisms: assault, sports, motor vehicle collisions. Penetrating mechanisms: firearms, explosions, impalement; series differ as to which predominates.
- Roughly 80% of penetrating scrotal trauma involves neighbouring structures — thigh, penis, perineum, bladder, urethra or femoral vessels.
- Genital wounds form a larger share of urological injuries in contemporary military conflict, because of powerful explosive weapons combined with the absence of body armour over the genitalia.
- Blast injury has a characteristic triad: extensive scrotal skin loss, multiple projectile injuries to both testes, and extensive destruction of the lower limbs and abdomen.

Diagnosis

- Testicular rupture must be considered in every case of blunt scrotal trauma. Typical presentation is exquisite scrotal pain with nausea.
- Swelling and ecchymosis are variable: the degree of haematoma does not correlate with the severity of injury and its absence does not exclude rupture, while contusion without fracture can itself produce significant bleeding.
- Scrotal haemorrhage, haematocoele and tenderness frequently prevent an adequate physical examination.
- Suspect and evaluate concomitant urethral injury where there is blood at the meatus, a suspicious mechanism, or haematuria. Penetrating injury mandates careful assessment of surrounding structures, especially the femoral vessels.
- Ultrasonography assesses testicular integrity and vascularity in equivocal cases — rapid, readily available and non-invasive.
- In blunt scrotal trauma its sensitivity and specificity for rupture are both modest, of the order of 64% to 75%, with only moderate to strong correlation between ultrasound and intraoperative findings.
- In scrotal gunshot wounds: sensitivity 60%, specificity 95% — ultrasound is insufficient as a screening tool to exclude testicular injury in penetrating trauma.
- Ultrasound features of testicular fracture: heterogeneous parenchymal echo pattern and disruption of the tunica albuginea.
- A normal or equivocal ultrasound must not delay exploration when examination suggests testicular damage — definitive diagnosis is often made in theatre (AUA Urotrauma Guidelines, Statement 30).
- MRI demonstrates testicular integrity effectively but is not in routine use — expense, limited availability, and the risk of delaying definitive surgery.
- Differential of testicular fracture: haematocoele without rupture, torsion of the testis or an appendage, reactive hydrocele, haematoma of the epididymis or spermatic cord, and intratesticular haematoma.
- A non-palpable testis in a trauma patient should raise the possibility of dislocation outside the scrotum. Dislocation classically follows motorcycle crashes, where extreme force expels the testis into surrounding tissue.
- Commonest destination: the superficial inguinal pouch (50%); other sites are pubic, penile, pelvic, abdominal or perineal. Bilateral dislocation is reported. Manual or surgical reduction is indicated.

- **Torsion must be considered in all cases of significant scrotal pain without signs of major scrotal trauma** — trauma does not exclude torsion.

Management

- **Early exploration and repair improve testicular salvage, shorten convalescence and disability, hasten return to normal activity, and preserve fertility and hormonal function.**
- **Minor scrotal injury without testicular damage** is managed with **ice, elevation, analgesia, and irrigation with closure where appropriate.**
- **Four objectives of exploration: testicular salvage, prevention of infection, control of bleeding, and reduced convalescence.** Either a transverse or a vertical scrotal incision may be used.
- **Repair as a sequence: remove necrotic and extruded seminiferous tubules → close the tunica albuginea with small absorbable sutures.**
- **Even small tunical defects are closed primarily** — progressive swelling and rising intratesticular pressure continue to extrude seminiferous tubules over time.
- **Every attempt at salvage should be made; loss of capsule may require removal of additional parenchyma to permit closure of the remaining tunica albuginea.**
- **A tunica vaginalis flap or graft covers a large tunical defect in an otherwise salvageable testis; synthetic grafts are not recommended.**
- **Significant intratesticular haematoma should be explored and drained even without rupture** — otherwise **progressive pressure necrosis and atrophy follow**, with substantial rates of **delayed exploration (40%) and orchidectomy (15%).**
- **Significant haematocele should be explored regardless of imaging** — up to 80% are caused by testicular rupture.
- **Penetrating scrotal injury should be explored to inspect for vascular and vasal injury; the same principles of salvage, haemostasis and reconstruction apply as in blunt trauma.**
- **Vas deferens injury → ligate with non-absorbable suture, with delayed reconstruction if required.** Consider exploration of the contralateral testis depending on mechanism and projectile trajectory.

Outcome and complications

- **Non-operative management of testicular rupture is frequently complicated by infection, atrophy, necrosis, chronic unrelenting pain and delayed orchidectomy.**
- **Testicular salvage exceeds 90% with exploration and repair within 3 days of injury; beyond 72 hours the salvage rate may fall dramatically.**
- **Blunt testicular rupture carries very high salvage rates. Penetrating testicular trauma was historically associated with salvage of only 32–65%, but contemporary civilian and combat series report salvage as high as 86%, attributed principally to heightened provider awareness and earlier exploration.**
- **Most operated patients retain adequate hormonal and fertility function.**
- **Testicular injury is exceedingly rare in individual and team contact sports and recreational activity.**
- **Counselling for the solitary testis: advise parents, recommend a protective cup, but state plainly that the benefits of continued physical activity far outweigh the potential risk.**
- **The American Academy of Pediatrics Committee on Sports Medicine and Fitness recommends that children with a solitary testicle be permitted universal participation in both non-contact and full-contact sports.**

Genital Skin Loss

Aetiology

- **Fournier gangrene** — necrotising gangrene from polymicrobial genital infection — is a common cause of extensive genital skin loss. The skin loss is iatrogenic, produced by the necessary acute debridement of necrotic skin to control infection.
- Other mechanisms: traction by mechanical devices (farm or industrial machinery), suction devices (vacuum cleaners), and burns.
- **Penile skin avulses** cleanly because the superficial penile tissue is loose areolar tissue — it is frequently torn free without damage to the underlying structures.
- **Significant scrotal skin loss** from penetrating trauma is uncommon, but is characteristic of explosive injury.
- **Penile burns** are rare but usually full-thickness, because the penile skin is so thin.
- Constricting bands may cause significant skin loss, but the commoner injury is direct pressure necrosis beneath the band, which usually heals with device removal alone.

Diagnosis and initial management

- **Cellulitis and Fournier gangrene** both present with genital oedema and erythema; the discriminators are that skin ischaemia with crepitus is the hallmark of Fournier gangrene and loss of scrotal rugae is highly suggestive of tissue necrosis.
- **Scrotal ultrasonography** and computed tomography may demonstrate subcutaneous air, a useful indicator of necrotising infection.
- Management rests on three pillars: parenteral antibiotics, resuscitation and surgical debridement.
- **Multiple debridements** of ischaemic or frankly necrotic skin are required over several days until active infection is controlled (AUA Urotrauma Guidelines, Statement 31).
- **Interim wound care:** frequent wet-to-dry dressing changes or vacuum-assisted closure (negative pressure wound therapy, NPWT) until planned primary coverage.
- **Inspection by the surgical team** at least daily is prudent — the hazard of NPWT is that it removes the daily look at the wound.
- **Early suprapubic urinary diversion** should be strongly considered in extensive injury, both to simplify wound care and to prevent urethral complications of prolonged catheterisation.
- **Hyperbaric oxygen** is advocated by some but not recommended by the authors — considerable expense and logistical complexity without proven benefit.
- **NPWT** may be used in complex Fournier gangrene with large wound surface areas to reduce mortality and, theoretically, promote granulation.
- **Genital burns** are treated as other burns: early resection of burn eschar and coverage with split-thickness skin graft (STSG) where possible; partial-thickness skin loss may be treated with silver sulfadiazine cream.
- **Deep penile electrical burns** warrant a conservative approach — the ultimate outcome is usually autopenectomy, death, or both, owing to the extent of concurrent injuries.

Penile reconstruction

- Options span secondary wound healing, primary closure, split-thickness skin grafting and flaps.
- **Selected uncircumcised patients:** mobilisation of redundant foreskin permits primary closure of middle-to-distal penile skin loss.

- **Scrotal rotation flaps** serve more proximal defects with limited skin loss, but the hair-bearing nature of scrotal skin risks an unacceptable cosmetic result.
- **Local abdominal and thigh flaps** are usable but cosmetically inferior to split-thickness skin grafting. Do not resurface with avulsed skin — it frequently becomes necrotic.
- **Preferred graft for the penis:** thick (0.012–0.015 inch), **NON-meshed split-thickness skin graft**, usually harvested from the anterior thigh with a pneumatic dermatome. Meshed grafts contract and are cosmetically inferior on the penis.
- **Any subcoronal skin** remaining after debridement must be excised before grafting — retained distal foreskin obstructs lymphatics and produces circumferential lymphoedema.
- **Timing:** wait 5 to 7 days after the last debridement to allow an adequate wound bed, or apply the criteria of at least 72 hours of culture-directed antibiotics, nutritional optimisation, visible granulation tissue, and normalisation of the white blood cell count.
- **Immediate graft stabilisation:** tie-over bolster or circumferential vacuum dressing.
- **Grafts on the penile shaft** never regain normal sensation; sexual function is nonetheless often preserved, because glans sensation is intact and corporal anatomy and blood flow are unaltered.

Scrotal reconstruction

- **Scrotal skin loss of up to 50%** can often be closed primarily.
- For extensive loss, temporising options are **thigh pouches, wet dressings, or vacuum pressure dressings**. Thigh pouches are largely historical and are not recommended initially — they risk transmitting the infective process into uninvolved tissue.
- **Reconstruction as a sequence:** mobilise local skin flaps first to cover as much of the defect as possible → cover the remainder with meshed split-thickness skin graft.
- **Split-thickness skin graft** is the commonest graft used for scrotal reconstruction.
- **On the scrotum** meshing is an advantage — it allows exudate to escape through the interstices and improves graft take, with a good cosmetic result.
- **Sew the spermatic cords and testes** together in multiple places before grafting to prevent a bifid neoscrotum.
- The neoscrotum appears unnaturally tight initially; the testes assume a more natural dependent position after 6 to 12 months.
- **Regional and free flaps** are alternatives. Fibrin sealant is useful as a tissue glue to promote healing and reduce drainage in complex genital reconstruction.

KEY POINTS: GENITAL TRAUMA

- **Penile fracture** is diagnosed on history and physical examination; penile ultrasonography is readily available, inexpensive and rapid in equivocal cases.
- **Gunshot wounds** to the penis should be repaired primarily, to prevent deformity and erectile dysfunction.
- **Animal bites** → irrigate, debride and close primarily. Most human bites → leave open.
- **Blunt scrotal injury** may be evaluated by ultrasonography for testicular rupture.
- **Penetrating scrotal injury** should undergo surgical exploration to assess and repair deep tissues and remove contamination.

BLADDER INJURY

Aetiology

- **The bladder is normally protected by its deep position within the bony pelvis** — injury therefore implies substantial force or a full bladder.
- **Most blunt bladder injuries follow rapid-deceleration motor vehicle collisions; also falls, crush injuries, assault and blows to the lower abdomen.**
- Two mechanisms operate with pelvic disruption: **the bony pelvis tears the bladder at its fascial attachments**, and **bone fragments may directly lacerate the organ.**
- Other causes: **penetrating trauma, iatrogenic surgical complication, and spontaneous rupture in patients with neuropathic disease, pre-existing bladder disease or prior urological surgery.**
- **Blunt bladder injury is rarely isolated** — **80–94% have significant associated non-urological injuries. Mortality (8–44%) is driven by those associated injuries, not by the bladder injury itself.**
- The reciprocal figures are the classic pairing: **pelvic fracture accompanies 83–95% of bladder injuries** (the commonest associated injury), yet **bladder injury occurs in only 5–10% of pelvic fractures.**
- **Sudden force on a full bladder raises intravesical pressure abruptly and may rupture the bladder without any pelvic fracture** — **characteristically at the dome.**
- **Penetrating bladder trauma accounts for 15–51% of bladder injuries**, usually **stab or gunshot wounds**, and likewise carries significant associated injury and mortality.
- **Obstetric and gynaecological procedures are the commonest cause of iatrogenic bladder injury during open surgery**; urological, general and orthopaedic operations also contribute.

Clinical Signs and Symptoms

- **Gross haematuria is the most reliable indicator and is present in nearly all cases.**
- Conscious patients report **non-specific suprapubic pain with inability to void.** Other signs: **lower abdominal bruising, muscle guarding and rigidity, diminished bowel sounds.**
- **Associated abdominal and pelvic injuries may mask bladder symptoms; raise suspicion further in the unresponsive patient** — **intoxication or altered sensorium.**

KEY POINTS: CLINICAL INDICATORS OF BLADDER INJURY

- **Suprapubic pain or tenderness**
- **Free intraperitoneal fluid on computed tomography (CT) or ultrasonography**
- **Inability to void, or low urine output**
- **Clots in the urine, or clots seen within the bladder on CT**
- **Enlarged scrotum with ecchymosis**
- **Abdominal distension or ileus**

Radiographic Imaging

- The anatomical correlation to hold: **extraperitoneal injury is usually associated with pelvic fracture**, whereas **intraperitoneal injury may accompany pelvic fracture but more commonly results from penetrating injury or a burst injury at the dome from a direct blow to a full bladder.**
- **Imaging matters because it dictates management** — the extraperitoneal versus intraperitoneal distinction determines operative versus non-operative treatment.

- **Absolute indication for immediate cystography after blunt trauma: gross haematuria with pelvic fracture** — approximately 29% of this group have bladder rupture (AUA Urotrauma Guidelines, Statement 14a).
- **Relative indications after blunt trauma: gross haematuria without pelvic fracture, and microscopic haematuria with pelvic fracture.** The yield in these atypical groups is very low — 0.6% for pelvic fracture with microscopic haematuria — so the threshold to image is set by associated clinical indicators of bladder injury.
- **Penetrating injury of the buttock, pelvis or lower abdomen with any degree of haematuria warrants cystography** — the threshold for penetrating trauma is far lower than for blunt.
- **Retrograde (stress) cystography is nearly 100% accurate if performed correctly. Fill the bladder to a sense of discomfort in the cooperative, conscious patient; otherwise to 350 mL.**
- Plain-film technique is a sequence of three images: scout film before contrast → full-bladder anteroposterior film → drainage (post-void) film. The drainage film is not optional — posterior extravasation is missed without it.
- **Significant bladder distension is required to demonstrate small tears** — an underfilled study is a false-negative study.
- **Upper and lower urinary tract injuries are almost never coincident (0.4%)** — haematuria and mechanism may still mandate upper tract imaging, but the two rarely coexist.
- **Extraperitoneal extravasation: a dense, flame-shaped collection of contrast within the pelvis.** Depending on fascial integrity, contrast may extend beyond the pelvis into the retroperitoneum, scrotum, phallus, thigh or abdominal wall. The volume of extravasation is not proportional to the size of the injury.
- **Intraperitoneal extravasation: contrast outlining loops of bowel and/or the lower lateral peritoneal cavity.**
- **CT is now routine in trauma, so concomitant CT cystography is frequently the more efficient means of assessing the bladder, and is as accurate and reliable as plain-film cystography, provided the bladder is filled retrogradely with contrast diluted to 2–4% (a 6:1 dilution with saline) to a volume of 350 mL.**
- **Dilution is mandatory** — undiluted contrast is so dense that scatter artefact degrades image quality.
- **Drainage films are not required after CT cystography, because the retrovesical space is well seen on axial images** — the deliberate contrast with plain-film technique, where the drainage film is obligatory.
- **Clamping the urethral catheter to allow antegrade distension by intravenously administered contrast is inadequate for diagnosing bladder rupture** — retrograde filling is mandatory.
- **Conventional abdominal CT may show findings suggestive of bladder injury but is not adequate for bladder evaluation without retrograde distension.**

Management

- **Uncomplicated extraperitoneal rupture is managed conservatively with urethral catheter drainage alone (AUA Urotrauma Guidelines, Statement 16).**
- **Use a large-bore 22-French Foley catheter to ensure adequate drainage. If output is poor, perform fluoroscopic cystography to confirm correct catheter position** — poor drainage is a catheter problem until proven otherwise.
- **Cystography must confirm complete healing before catheter removal at 2 to 3 weeks. If the injury persists beyond 4 weeks of catheter drainage, consider surgical repair.**
- **Bone spicules within the bladder wall may impair healing and are themselves an indication for repair.**

- **Contemporary comparative data favour cystorrhaphy over conservative management: fewer urological complications such as fistula, shorter hospital stay and quicker time to a negative cystogram. Blunt extraperitoneal injury with complicating features therefore warrants immediate open repair, to prevent fistula, abscess and prolonged leak.**
- **If the patient is undergoing laparotomy for other injuries, repair the extraperitoneal rupture at the same sitting. Technique as a sequence: enter the anterior bladder wall directly in the midline → close the tear intravesically with absorbable suture. Do not disturb the perivesical pelvic haematoma.**
- **Data on whether an extraperitoneal injury must be repaired when internal fixation of the pelvic fracture is performed are limited; available evidence did not show a significant hardware infection rate when the bladder injury was left unrepaired, but the data are few and at high risk of bias.**
- **Penetrating and intraperitoneal injuries from external trauma require immediate operative repair; operative repair has been associated with a 59% reduction in mortality.**
- **The rationale is twofold: these injuries are often larger than cystography suggests, and they are unlikely to heal spontaneously. Prolonged urinary leak causes chemical peritonitis or abscess.**
- **Selected patients with isolated intraperitoneal injury may undergo laparoscopic repair.**
- **When the bladder is explored after penetrating trauma without preliminary imaging, inspect the ureteric orifices for clear efflux; integrity may also be confirmed by intravenous indigo carmine, methylene blue or fluorescein green, or by retrograde passage of a ureteric catheter.**
- **Penetrating injury involving the ureteric orifice or intramural ureter → primary closure with stented ureteric reimplantation and a perivesical drain.**
- **Antimicrobials for 3 days perioperatively only in intraperitoneal rupture. After repair, obtain a cystogram at 7 to 10 days.**
- **Suprapubic tube drainage confers no benefit over urethral catheter drainage alone — catheter drainage alone is sufficient (AUA Urotrauma Guidelines, Statement 18).**
- **Concurrent rectal or vaginal injury, as a sequence of principles: separate the organ walls → avoid overlapping suture lines → interpose viable tissue between the repaired structures. Fibrin sealant over the bladder closure may reduce complications where no interposition tissue is available.**

KEY POINTS: INDICATIONS FOR IMMEDIATE REPAIR OF BLADDER INJURY

- **Intraperitoneal injury**
- **Penetrating injury**
- **Iatrogenic urological injury (stable patient)**
- **Inadequate bladder drainage, or clots in the urine**
- **Bladder neck injury**
- **Rectal or vaginal injury**
- **Selected stable patients undergoing laparotomy for other reasons**
- **Bone fragments projecting into the bladder**

Outcomes and Complications

- **Prompt diagnosis and appropriate management give excellent results with minimal morbidity.**
- **Serious complications follow delayed diagnosis or treatment — through misdiagnosis, delayed presentation, or complex injury from devastating pelvic trauma.**
- **Unrecognised bladder injury presents as acidosis, azotaemia, fever and sepsis, low urine output, peritonitis, ileus, urinary ascites, or respiratory difficulty. The mechanism worth carrying: intraperitoneal**

urine is reabsorbed across the peritoneum, producing acidosis and azotaemia in a patient with normal renal function.

- Unrecognised bladder neck, vaginal and rectal injuries produce incontinence, fistula, stricture and difficult delayed major reconstruction.
- Severe pelvic fracture may cause transient or permanent neurological injury, with voiding difficulty despite an adequate bladder repair — a failure of function that is not a failure of the repair.

URETHRAL INJURY

Posterior Urethral Injuries

Aetiology

- Pelvic fracture urethral injury (PFUI) typically occurs as part of multisystem trauma from motor vehicle collisions, falls or industrial accidents.
- Fracture of the anterior pelvic ring or pubic symphysis diastasis is usually present when urethral disruption occurs, and a greater degree of displacement correlates with a higher risk of urethral injury.
- Highest-risk patterns: straddle fractures involving all four pubic rami, and fractures producing vertical and rotational pelvic instability.
- Incidence with pelvic fracture: approximately 10% of males, up to 6% of females.
- The posterior urethra is densely adherent to the pubis via the urogenital diaphragm and the puboprostatic ligaments — this fixation is what converts pelvic displacement into urethral disruption.
- The bulbomembranous junction, not the prostatomembranous junction, is the vulnerable site.
- Endoscopic and urodynamic evaluation confirms that the membranous urethral sphincter complex usually remains functionally intact, being avulsed vertically, posteriorly or laterally from the underlying bulb — hence continence is often preserved after reconstruction.

Diagnosis — examination

- Classic triad of urethral disruption: inability to urinate, blood at the meatus, and a palpably full bladder.
- The classic signs are frequently absent or hard to elicit — including the "high-riding" prostate and the "butterfly" perineal haematoma.
- In practice, disruption is often first recognised when the trauma team cannot pass a urethral catheter, or when the catheter is misplaced into the pelvic haematoma.
- Pelvic haematoma obscures the prostatic contour and produces the false impression of an impalpable, high-riding prostate.
- Females may sustain proximal urethral avulsion, presenting with vulvar oedema and blood at the vaginal introitus — careful vaginal examination is mandatory in every female with a pelvic fracture.

Diagnosis — urethrography

- Blood at the urethral meatus mandates immediate retrograde urethrography (AUA Urotrauma Guidelines, Statement 19).
- Technique as a sequence: place a small-bore 16-French catheter unlubricated 1 cm into the fossa navicularis → inflate the balloon with 3 mL of water for a snug fit → apply penile traction with a Brodney clamp or rolled gauze → position obliquely or in lateral decubitus → inject 20 mL of contrast gently → expose the film during injection.
- A catheter tip attached to a Luer-Lok syringe filled with contrast is an alternative delivery method. Fluoroscopy is preferable where available.

- Oblique positioning may be impossible in trauma; the patient may remain supine.
- In females, urethrography is difficult and may not demonstrate the injury — thorough physical examination and urethroscopy are the assessment of choice.

Initial management

- Injuries are classified as major distraction, minor distraction, and partial urethral injury; the discriminator between major and minor distraction is whether a cystoscope can be navigated into the bladder retrogradely.
- Injury severity predicts both outcome and management — rates of urethral obstruction and of eventual urethroplasty fall progressively from major distraction, to minor distraction, to partial injury.
- Immediate suprapubic tube placement remains the initial standard of care in males with posterior urethral injury. Percutaneous placement is safe and expedient when the bladder is obviously distended and no other operative indication exists.
- Technique as a sequence: Trendelenburg position → preliminary 18-gauge spinal needle → peel-away sheath trocar in the lower midline → 16-French Foley catheter.
- Avoid smaller "punch" suprapubic tubes — less robust, prone to fracture or obstruction with debris, requiring urgent replacement within weeks.
- Alternative: infraumbilical incision, which permits direct inspection and repair of a concomitant bladder injury and placement of a large-bore catheter at the bladder dome. Delayed urethral reconstruction then proceeds electively within a few months, once the patient is ambulant.
- Orthopaedic surgeons increasingly perform early pelvic ring fixation to reduce bleeding, improve healing and hasten ambulation. The examinable controversy: orthopaedic concern that a suprapubic tube may seed anterior pubic hardware versus urological evidence that suprapubic cystostomy is safe throughout the course of care (AUA Urotrauma Guidelines, Statement 21).
- Practical compromise when transurethral access is compromised and the fracture is managed by open reduction and internal fixation: place a large-bore Foley in the bladder, tunnelled through the skin as high on the lower abdominal midline as possible, to keep the tube away from the plated symphysis.
- A brief attempt at primary realignment is a recognised alternative in the stable patient (AUA Urotrauma Guidelines, Statement 22), performed acutely or within several days, using combined retrograde and antegrade flexible cystoscopy.
- Prolonged endoscopic realignment attempts risk infection and are not recommended — the emphasis is on *brief*.
- Prospective multi-institutional data show no statistically significant difference between suprapubic catheter placement and early endoscopic realignment in rates of urethral obstruction or need for urethroplasty. Where simple retrograde cystoscopy fails to place a urethral catheter, suprapubic catheter placement is the preferred early option.
- Post-realignment protocol as a sequence: remove the urethral catheter at 4 to 6 weeks → retain the suprapubic catheter, because the overwhelming majority develop posterior urethral stenosis despite realignment → if voiding per urethra is satisfactory, remove the suprapubic catheter 7 to 14 days later.
- Primary realignment rarely heals without stricture. Whether realigned or not, a urethral stenosis of 1 to 2 cm typically develops in the overwhelming majority, requiring elective delayed posterior urethroplasty.
- Repeated endoscopic treatment or dilatation of PFUI is costly, prone to failure, and adds complexity to the eventual repair.

- Cost-effectiveness modelling: **primary endoscopic realignment followed by a single attempted direct visual internal urethrotomy (DVIU) before definitive urethroplasty was most cost-effective — but only if the injury is not significant, is amenable to realignment, and DVIU success is at least 32%; below that threshold, suprapubic tube with delayed urethroplasty is the most cost-effective strategy.**
- **Incomplete tears are best treated by stenting with a urethral catheter. The concern that catheterisation converts a partial tear into a complete injury is not supported by evidence — a gentle attempt at catheterisation is acceptable, though misplacement outside the bladder remains possible and radiographic confirmation of position is imperative.**

Delayed reconstruction

- **The rupture defect between the severed ends fills with scar, producing complete loss of urethral continuity. This is not a stricture — it is a true urethral stenosis filled with fibrosis.**
- **At 2 to 3 months the scar is stable enough for safe posterior urethroplasty, provided associated injuries are stabilised and the patient is ambulant. Maintain suprapubic drainage until then.**
- **Obtain a cystogram and retrograde urethrogram simultaneously — the "up-and-down-o-gram".**
Technique: the patient attempts to void with the bladder filled → ideally the prostatic urethra is visualised as the bladder neck opens → the distance between the severed ends is measured.
- **If the bladder neck does not open, supplement with flexible endoscopy.**
- **The radiographic appearance of the bladder neck does not reliably predict its postoperative behaviour — incompetence or obstruction cannot be forecast from imaging.**
- **MRI defines defect length, the extent and direction of urethral dislocation, and the degree of prostatic displacement, and may assist in planning the approach.**

Endoscopic treatment

- **DVIU is best reserved for selected short, incomplete stenoses in which early catheterisation preserved urethral continuity.**
- **Defects of 1 cm or longer are not amenable to endoscopic treatment — "cut-to-the-light" procedures, which cut through pelvic scar to create a channel between the avulsed ends, tend to be ineffective at this length.**
- **Aggressive endoscopic treatment risks coring a false passage that inadvertently bypasses the bladder neck.**
- **Core-through procedures commit the patient to repeated urethrotomies or long-term dilatation, and the fibrosis inevitably contracts, producing difficult self-catheterisation, false passage and acute urinary retention.**
- **In such cases, a 2- to 3-month period of "urethral rest" on suprapubic diversion is advised before open reconstruction.**

Surgical reconstruction

- **Open posterior urethroplasty by a perineal anastomotic approach is the treatment of choice for most urethral distraction injuries — it cures definitively without multiple procedures. Prerequisite: preoperative studies must show the prostatic apex is reachable perineally.**
- **Technique as a sequence: high lithotomy position → midline or lambda-shaped incision → free and mobilise the bulbar urethra from the rupture site to the penoscrotal junction → completely excise the scar of the rupture defect → identify the prostatic urethra at the apex → meticulously excise all fibrotic tissue from the proximal urethral margin → tension-free anastomosis of bulbar to prostatic urethra.**
- **A perineal approach alone achieves anastomosis in the majority of cases.**

- **Adjunctive manoeuvres when direct anastomosis is difficult, in escalating order: corporal splitting, inferior pubectomy, corporal rerouting.** Several groups find these manoeuvres, particularly supracrural rerouting, unnecessary or unhelpful in many cases.
- **Total removal of the symphysis pubis (Pierce, 1962)** is recommended for **severe injury with complicating features — fistula, marked displacement, or retropubic fixation of the prostate.**
- **Combined abdominoperineal approach, with or without partial pubectomy,** is useful in **severe fibrosis, fistula, previous failed anastomotic urethroplasty, associated bladder neck injury, and paediatric cases.**
- **Limit lithotomy time to 5 hours or less in any complex urethral reconstruction, to prevent lower limb complications.**

Complications

- **Some degree of impotence occurs in as many as 82% of patients with pelvic fracture and urethral distraction injury.**
- **Aetiology is multifactorial: cavernous nerve injury, arterial insufficiency, venous leak, and direct corporal injury.**
- **Predictors track injury severity: pubic symphyseal diastasis, lateral displacement of the urethra, grade of urethral injury, and a long urethral gap.**
- **ED and incontinence are usually consequences of the injury itself, not of the treatment. Reports of higher ED and incontinence with primary realignment predate flexible endoscopy, and with modern primary endoscopic realignment, impotence and incontinence rates match those of untreated patients and of delayed open reconstruction.**
- **Small numbers in surgical series show new or worsened ED after reconstruction, but injury effects cannot be separated from repair effects.**
- **Other sexual complications to counsel on: ejaculatory dysfunction, change in penile length, and penile curvature.**
- **Arterial insufficiency is present in many men rendered impotent by pelvic fracture. Impotent patients may be more vulnerable to restenosis after posterior urethroplasty, through bulbar urethral ischaemia, so some experts advise preoperative penile arterial duplex Doppler in "at-risk" patients to identify candidates for initial penile revascularisation. Indications for penile revascularisation in post-traumatic ED are nonetheless extremely limited.**
- **Recurrent stenosis at the anastomosis after posterior urethroplasty: 5–15%. DVIU is often successful here, because most fibrotic tissue has already been excised — the deliberate contrast with DVIU on an untreated distraction defect, which fails.**
- **Incontinence after reconstruction is low — less than 4%. Continence is thought to depend largely on bladder neck function, though a significant proportion retain identifiable distal rhabdosphincter function on urodynamics.**
- **Monitor for incontinence and ED for at least 1 year after urethral injury (AUA Urotrauma Guidelines, Statement 23).**

Anterior Urethral Injuries

Aetiology

- **Most anterior urethral injuries follow straddle injury and involve the bulbar urethra, which is susceptible to compression because of its fixed position beneath the pubis.**
- **A smaller proportion result from penetrating penile injury, foreign body insertion, or iatrogenic trauma.**

- Maintain a high index of suspicion in all blunt or penetrating urogenital trauma; perform urethrography in any suspected urethral injury.
- Clinical signs: blood at the meatus, perineal haematoma, gross haematuria, urinary retention.
- Establish prompt urinary drainage by suprapubic tube or urethral catheterisation (AUA Urotrauma Guidelines, Statement 25).
- If Buck's fascia is disrupted in severe trauma, blood and urine extravasate into the scrotum — the same fascial principle as in penile fracture.
- The principal morbidity of straddle injury is urethral stricture, which may declare itself years later.

Initial management

- Armenakas and McAninch classification, based on radiographic findings: contusion, incomplete disruption, complete disruption.
- Contusion and incomplete disruption → urethral catheter diversion alone.
- Major straddle injury involving the urethra → initial suprapubic cystostomy is the standard of care; primary anterior realignment shows promising stricture and ED results in straddle injuries of lesser magnitude.
- Low-velocity urethral gunshot injury → primary surgical repair, since catheter alignment alone carries a far worse stricture rate. High-velocity gunshot wound → initial suprapubic diversion with delayed reconstruction.
- Limit debridement of the corpus spongiosum — its blood supply is robust and most contused areas heal spontaneously.

Delayed reconstruction

- Obtain a retrograde urethrogram and voiding cystourethrogram before any planned procedure, to define the site and length of the obliterated segment.
- Urethral ultrasonography delineates stricture length and severity: retrograde saline injection combined with antegrade bladder filling distends the urethra proximally and distally, and a 10-MHz sonogram defines the non-distensible segment for excision. Post-traumatic dense fibrosis appears fixed and non-distensible with significant acoustic shadowing.
- Anastomotic urethroplasty is the procedure of choice for the totally obliterated bulbar urethra after straddle injury. The typical scar is 1.5 to 2 cm and can be completely excised, allowing mobilisation of proximal and distal urethra for a tension-free end-to-end anastomosis. Success exceeds 95%.
- Endoscopic incision through the scar of an obliterated urethra after straddle injury is futile. Partial urethral narrowing, by contrast, can be treated initially by endoscopic incision or dilatation with greater success.
- Repeated endoscopic manipulation is neither clinically nor cost-effective, and patients so treated are more likely to require complex reconstruction with grafts.
- Delay open repair for several weeks after instrumentation to allow the urethra to stabilise; a 2-month period of suprapubic diversion may be prudent before repair of complex or recurrent catheter-dependent strictures.

KEY POINTS: LOWER GENITOURINARY TRACT INJURIES

- Most bladder injuries are associated with pelvic fracture, but only about 10% of patients with pelvic fracture have a bladder injury.
- Intraperitoneal bladder injury from external trauma requires urgent operative repair.

- **Prompt suprapubic catheter drainage is recommended for initial management of pelvic fracture urethral disruption injury.**
- **Primary realignment may be performed in stable patients; close follow-up is essential because of the high rate of stricture formation.**
- **Posterior urethroplasty by excision and primary anastomosis is the treatment of choice for delayed management of PFUI.**
- **Anastomotic urethroplasty is the procedure of choice for straddle injury-related anterior urethral stricture.**